

»» Chapter XII ««

TOPICS IN THE THEORY OF POWER SERIES

PREFATORY REMARKS

We have seen in Ch. IV that convergent power series enjoy many properties useful from a formal point of view. We recall the convenient rules for the calculation of products and derivatives which make power series such valuable tools in the study of differential equations. Simple applications of these facts were given in Ch. IV. Our object in the present chapter is to extend our knowledge of power series in various directions. For one thing it is worth our while to systematize the technique of undetermined coefficients which was used earlier in an *ad hoc* exploratory manner. We are led to one of Cauchy's many fundamental contributions—the majorant calculus. With no greater effort than that involved in the most elementary of situations we may employ the same techniques of Cauchy to study linear differential equations with solutions taking values in a Banach space.† Other topics treated will be: formal aspects of power series and Laurent series for which convergence plays no part; calculation with power series; holomorphic functions in two variables; and, in the final section, the notion of analytic continuation of a power series—a concept endowing the Weierstrassian vision of complex function theory with added harmony and structural insight.

In the final chapter of this book we shall introduce the Riemannian point of view and we shall indicate in an important case (the problem of Riemann and Klein) how the notion of analytic continuation or, better still, a mild

† The appropriate definitions will be given below.

extension serves as a bridge between the classical theory of analytic functions, which is the dominant theme of this book, and the modern theory of abstract Riemann surfaces.

The theory of power series is vast and it is apparent to the experienced worker that very many important ideas and developments of the subject are not even touched in this book. The refined theory of power series is most appropriately treated in the framework of a specialized monograph rather than in that of a general introductory text. The bibliographical indications will serve as a guide to the interested reader.

This chapter presupposes familiarity with Chs. I–VII inclusive as well as the material on sums in Ch. VIII.

1. FORMAL ASPECTS

The first three problems of the exercise set below underscore algebraic aspects of the theory of power series and suitably restricted Laurent series. A basic tool is the notion of Cauchy multiplication, which we have seen arises inevitably in the theory of power series with radius of convergence greater than zero.

EXERCISES

1.1. Let F denote a field. Let A denote the set of sequences $(a_n): \mathbf{N} \rightarrow F$. We define an addition on A by pointwise addition,

$$(a_n) + (b_n) = (a_n + b_n),$$

and a multiplication (Cauchy multiplication) by

$$(a_n) \cdot (b_n) = \left(\sum_{j+k=n} a_j b_k \right).$$

Here the notation $\sum_{j+k=n} a_j b_k$ means the sum over the (j, k) with $j + k = n$. This notation will be used in the next exercise where infinite index sets enter and *sum* is construed for a map f into an abelian group G for which $f^{-1}(G - \{0\})$ is finite as the usual sum of f restricted to $f^{-1}(G - \{0\})$. Show that A together with addition and multiplication so defined is a commutative ring with identity having no zero divisors (domain of integrity). Show that $D((a_n)) = ((n+1)a_{n+1})$ defines a map of A into itself which has the properties of a derivation; that is, if $a, b \in A$, then

$$D(a+b) = D(a) + D(b) \quad \text{and} \quad D(ab) = D(a)b + aD(b).$$

[The inductive definition of na , $n \in \mathbf{N}$, $a \in F$, specified by $0a = 0$, $(n+1)a = na + a$, is to be recalled.]

1.2. Let B denote the set of sequences $(a_n): \mathbf{Z} \rightarrow F$ (F as in Ex. 1.1), satisfying

$\inf\{n: a_n \neq 0\} > -\infty$. Show that with definitions analogous to those for A of addition and multiplication B is a field.

1.3. Let $(a_n) \in A$ and $\varphi((a_n)) = (b_n) \in B$ where $b_n = 0$, $n < 0$, $b_n = a_n$, $n \in \mathbb{N}$. Show that φ is an isomorphism of A into B and that the subfield of B generated by $\varphi(A)$ is precisely B . What happens to the above results when $F = \mathbb{C}$ and A and B are taken as the sets of sequences $(a_n)_{n \in \mathbb{N}}$ and $(b_n)_{n \in \mathbb{Z}}$, respectively, such that $\sum_0^\infty a_n z^n$ and $\sum_0^\infty b_n z^n$ have radii of convergence greater than zero and $\inf\{n: b_n \neq 0\} > -\infty$?

1.4. If (a_n) is the 0-Laurent sequence of $f \in M(\hat{\mathbb{C}})$, there exists $(b_n): \mathbb{Z} \rightarrow \mathbb{C}$, $b_n = 0$ for $n < 0$, $b_0 = 1$, $b_n = 0$ for $n \geq m$ for some positive integer m such that the components of $(a_n)(b_n)$ with indices $\geq m$ are zero. Show that if $(a_n): \mathbb{Z} \rightarrow \mathbb{C}$ satisfies $\inf\{n: a_n \neq 0\} > -\infty$ and there exists (b_n) of the above kind such that $(a_n)(b_n)$ has the stated property, then (a_n) is the 0-Laurent sequence of some $f \in M(\hat{\mathbb{C}})$.

2. SUMS, CAUCHY MULTIPLICATION

EXERCISES

2.1. Let $f: X \rightarrow \mathbb{C}$ be summable. Suppose that $\mathfrak{P}$ is a collection of subsets of X satisfying $\bigcup_{Y \in \mathfrak{P}} Y = X$; if $Y, Z \in \mathfrak{P}$ and $Y \neq Z$, then $Y \cap Z = \emptyset$. We define $F(Y)$ as $s(f|Y)$. Show that F is summable and that

$$\sum_{Y \in \mathfrak{P}} F(Y) = \sum_{x \in X} f(x).$$

A special case was considered in Ex. 3.1, Ch. IX.

2.2. Let $f: X \rightarrow \mathbb{C}$ and $g: Y \rightarrow \mathbb{C}$ be summable. Let $h(x, y) = f(x)g(y)$ with domain $X \times Y$. Show that h is summable and that

$$\sum_{(x,y) \in X \times Y} h(x, y) = \left[\sum_{x \in X} f(x) \right] \left[\sum_{y \in Y} g(y) \right].$$

2.3. Apply Exs. 2.1 and 2.2 to the situation of absolutely convergent series $\sum_0^\infty a_n$, $\sum_0^\infty b_n$ to conclude that $\sum_0^\infty c_n$ is absolutely convergent and

$$\left(\sum_0^\infty a_n \right) \left(\sum_0^\infty b_n \right) = \sum_0^\infty c_n.$$

Throughout this exercise (c_n) is the product $(a_n)(b_n)$ in the sense of the preceding section.

Show that under the assumption that $\sum_0^\infty a_n$ is absolutely convergent and $\sum_0^\infty b_n$ is convergent, $\sum_0^\infty c_n$ is convergent and has sum $(\sum_0^\infty a_n)(\sum_0^\infty b_n)$. Show that the theorem is false if the hypothesis of absolute convergence is dropped for $\sum_0^\infty a_n$. It suffices to take $a_n = b_n = (-1)^n d_n$ where $(d_n)_{n \in \mathbb{N}}$ is a nowhere increasing sequence of positive numbers satisfying: $\lim_{n \rightarrow \infty} d_n = 0$, $\lim_{n \rightarrow \infty} d_n (\sum_0^n d_k) = +\infty$. Give an explicit instance of such a sequence (d_n) .

Show, however, that if both $\sum_0^\infty a_n$ and $\sum_0^\infty b_n$ are convergent, then

$$\lim_1 f = \left(\sum_0^\infty a_n \right) \left(\sum_0^\infty b_n \right),$$

f being $\sum_0^\infty c_n x^n$ with domain $[0, 1[$.

3. HOLOMORPHIC FUNCTIONS OF TWO COMPLEX VARIABLES

With the experience of one-variable complex function theory at our disposal, it is not difficult to treat the elementary aspects of the local theory of complex-valued functions of several complex variables. The difficulties of the several variable theory, which are serious, come at a later stage. The present section develops some of the simpler aspects of the two-variable theory starting from the power series point of view. A modern treatment of analytic functions of several complex variables is to be found in [48].

We start with (a_{mn}) : $\mathbb{N}^2 \rightarrow \mathbb{C}$ and study the set of $(z, w) \in \mathbb{C}^2$ for which $a_{mn} z^m w^n$, domain $\mathbb{N}^2$, is summable; that is, we are concerned with the summability set of the family of functions $a_{mn} z^m w^n$ with domain $\mathbb{C}^2$.† Suppose that $a_{mn} \alpha^m \beta^n$ is summable, $\alpha\beta \neq 0$. Then $|a_{mn} \alpha^m \beta^n|$ is summable and consequently $a_{mn} z^m w^n$ is uniformly summable on $\Delta(0; |\alpha|) \times \Delta(0; |\beta|)$.‡ It follows that Ω , the interior of the summability set of $a_{mn} z^m w^n$, is the union of a family of bicylinders $\Delta(0; r) \times \Delta(0; s)$. Further, at each point (Z, W) of the exterior of the summability set,

$$\sup\{|a_{mn} Z^m W^n|: (m, n) \in \mathbb{N}^2\} = +\infty.$$

Indeed, if $ZW \neq 0$ and $|a_{mn} Z^m W^n| \leq M < +\infty$, then

$$|a_{mn} z^m w^n| \leq M \left| \frac{z}{Z} \right|^m \left| \frac{w}{W} \right|^n,$$

and hence, for $|z| < |Z|$, $|w| < |W|$, the summability of $a_{mn} z^m w^n$ is assured, and hence (Z, W) is not an exterior point of the summability set. If $W = 0$, then there exists z , $|z| < |Z|$, such that $\sum_0^\infty a_{m0} z^m$ diverges and hence

$$\sup\{|a_{m0} Z^m|: m \in \mathbb{N}\} = +\infty;$$

the same reasoning holds when $Z = 0$.¶

The sum function f is continuous at each point of Ω . As in the one-variable theory, which will serve both as a guide and an aid, f enjoys differentiability properties and others as well.

We note that the interior of the summability set of $(m+1)a_{m+1,n} z^m w^n$ and

† From this point on X^n denotes the set of $(x_1, \dots, x_n)$, $x_k \in X$, $k = 1, \dots, n$. We shall omit such terms as “domain $\mathbb{N}^2$ ” when the matter is clear from context. We could, if we wished, introduce formally a language of power series in two variables. This is hardly worth our while for we have in the sum concept a convenient and adequate medium.

‡ A cartesian product of disks $\Delta(a; r) \times \Delta(b; s)$ is termed a *bicylinder*, and a product of the form $\{|z - a| \leq r, |w - b| \leq s\}$, $r, s \geq 0$, a *closed bicylinder*.

¶ We recall that an *exterior point* of a set X is an interior point of its complement, and that the *exterior* of X is the set of exterior points of X .

that of $(n+1)a_{m,n+1}z^m w^n$ contain Ω . It suffices to consider the first case, the argument for the second being the same. If $a_{mn}\alpha^m\beta^n$ is summable, $\alpha\beta \neq 0$, then from

$$\begin{aligned} |(m+1)a_{m+1,n}z^m w^n| &= |(m+1)a_{m+1,n}\alpha^m\beta^n| \left| \frac{z}{\alpha} \right|^m \left| \frac{w}{\beta} \right|^n \\ &\leq (m+1) \frac{M}{|\alpha|} \left| \frac{z}{\alpha} \right|^m \left| \frac{w}{\beta} \right|^n, \end{aligned}$$

$0 < M < +\infty$, we conclude that $\Delta(0; |\alpha|) \times \Delta(0; |\beta|)$ is contained in the summability set of $(m+1)a_{m+1,n}z^m w^n$. The asserted inclusion follows. That inclusion cannot be sharpened to equality follows from the example: $a_{mn} = 1$, $mn = 0$ and $a_{mn} = 0$, $mn \neq 0$. Here $\Omega = [\Delta(0; 1)]^2$ but for $(m+1)a_{m+1,n}z^m w^n$ the set in question is $\Delta(0; 1) \times \mathbb{C}$.

Suppose now that we fix w such that $(z, w) \in \Omega$ for some z . The set $\{z: (z, w) \in \Omega\}$ is clearly a disk $\Delta[0; \rho(w)]$, $\rho(w) > 0$, and for such z we have

$$f(z, w) = \sum_0^\infty g_k(w) z^k$$

where $g_k(w) = \sum_0^\infty a_{kj} w^j$, the series being absolutely convergent. We conclude that the partial derivative of f with respect to z at $(z, w) \in \Omega$ exists (the definition of partial derivative paraphrasing that of the real theory) and is given by

$$\sum_0^\infty (k+1) g_{k+1}(w) z^k.$$

A similar observation holds for the partial derivative of f with respect to w .

Let O be an open subset of $\mathbb{C}^2$ and let $f: O \rightarrow \mathbb{C}$. We say that f is *holomorphic on O* provided that for each $(\alpha, \beta) \in O$ there exist a bicylinder $\Delta(0; r) \times \Delta(0; s)$ and $(a_{mn}): \mathbb{N}^2 \rightarrow \mathbb{C}$ such that $a_{mn}z^m w^n$ is summable and has sum $f(z + \alpha, w + \beta)$ for $(z, w) \in \Delta(0; r) \times \Delta(0; s)$. The standard usages of this book concerning “at” and “in” are to be understood to prevail for the term holomorphic.

We are now ready to study some elementary facts concerning holomorphic functions in exercise form.

EXERCISES

The notation “ O ” will be reserved to denote an open subset of $\mathbb{C}^2$ in this exercise set.

3.1. Let $(\alpha, \beta) \in \mathbb{C}^2$, let U be a neighborhood of (α, β) , and let $f: U \rightarrow \mathbb{C}$. The definition of the differentiability of f at (α, β) paraphrases the definition of Ch. XI: f is

differentiable at (α, β) provided that there exist $A, B, C, D \in \mathbb{C}$ and $\varepsilon: U \rightarrow \mathbb{C}$, where $\varepsilon(\alpha, \beta) = 0$ and ε is continuous at (α, β) such that

$$f(z, w) - f(\alpha, \beta) = A(z - \alpha) + \overline{B(z - \alpha)} + C(w - \beta) + \overline{D(w - \beta)} + \varepsilon(z, w) (|z - \alpha|^2 + |w - \beta|^2)^{1/2}.$$

The coefficients A, B, C, D are unique. The usual notations $f_z, f_{\bar{z}}, f_w, f_{\bar{w}}$ will be used.

Now suppose that f is holomorphic on O . Show that f is differentiable at each point of O and that $f_{\bar{z}} = f_{\bar{w}} = 0$.

3.2. Reference is made to the definition of *holomorphic* given in the text. Let $0 < t < r, 0 < u < s$. Show that

$$a_{mn} t^m u^n = (2\pi i)^{-2} \int_0^{2\pi} \left[\int_0^{2\pi} f(\alpha + te^{i\theta}, \beta + ue^{i\varphi}) e^{-i(m\theta + n\varphi)} d\varphi \right] d\theta. \quad (3.1)$$

Let

$$M = \max\{|f(\alpha + te^{i\theta}, \beta + ue^{i\varphi})| : \theta, \varphi \in \mathbb{R}\}.$$

Show that $|a_{mn}| t^m u^n \leq M$ (Cauchy inequalities). Show with the aid of (3.1) that if

$$f(z + \alpha, w + \beta) = \sum_{(m,n) \in \mathbb{N}^2} b_{mn} z^m w^n$$

in $\Delta(0; r) \times \Delta(0; s)$, then $a_{mn} = b_{mn}$.

3.3. Show that the following Cauchy formulas hold for f in $\Delta(\alpha; t) \times \Delta(\beta; u)$, the setting being that of Ex. 3.2:

$$\begin{aligned} f(z, w) &= (2\pi i)^{-2} \oint_{C(\alpha; t)} \left[\oint_{C(\beta; u)} \frac{f(\zeta, \omega)}{(\zeta - z)(\omega - w)} d\omega \right] d\zeta \\ &= (2\pi i)^{-2} \oint_{C(\beta; u)} \left[\oint_{C(\alpha; t)} \frac{f(\zeta, \omega)}{(\zeta - z)(\omega - w)} d\zeta \right] d\omega, \end{aligned} \quad (3.2)$$

$(z, w) \in \Delta(\alpha; t) \times \Delta(\beta; u)$.

3.4. Develop the basic facts concerning repeated integrals of Cauchy type in two variables. Specifically, let γ and Γ be rectifiable paths in $\mathbb{C}$ each with domain $[0, 1]$, let g be a continuous map of $\gamma \times \Gamma$ (the product of the images) into $\mathbb{C}$. Show that

$$\int_{\gamma} \left[\int_{\Gamma} \frac{g(\zeta, \omega)}{(\zeta - z)(\omega - w)} d\omega \right] d\zeta$$

defines a holomorphic function f on $\mathbb{C}^2 - \gamma \times \Gamma$ and that for $(z, w) \in \mathbb{C}^2 - \gamma \times \Gamma$ the above integral is equal to

$$\int_{\Gamma} \left[\int_{\gamma} \frac{g(\zeta, \omega)}{(\zeta - z)(\omega - w)} d\zeta \right] d\omega.$$

Show that the derivatives of f of higher order are independent of the order of differentiation.

3.5. Given f holomorphic on O and t, u positive such that $\overline{\Delta(\alpha; t)} \times \overline{\Delta(\beta; u)} \subset O$. Show that the formulas (3.2) remain valid. Conclude that

$$f(z + \alpha, w + \beta) = \sum_{(m,n) \in \mathbb{N}^2} a_{mn} z^m w^n, \quad |z| < r, \quad |w| < s,$$

when $\Delta(\alpha; r) \times \Delta(\beta; s) \subset O$, the a_{mn} being independent of allowed (r, s) .

Show that if O is connected and $\text{int}\{(z, w): f(z, w) = 0\}$ is not empty, then f is the constant 0.

3.6. Conclude from Exs. 3.4 and 3.5 that if $(f_n)_{n \in \mathbb{N}}$ is a sequence of functions holomorphic on O tending uniformly in O (in the sense of the euclidean metric) to f , then f is holomorphic on O . In particular, the sum function f of $a_{mn} z^m w^n$ of the text is holomorphic in Ω .

3.7. Let f and g be holomorphic on O . Show that $f + g$ and fg are holomorphic on O and that f/g is also, provided that $g(z, w) \neq 0, (z, w) \in O$. Hint: For the product cf. Ex. 2.2, this chapter; for the quotient proceed locally with the aid of the geometric series:

$$[g(z, w)]^{-1} = [g(\alpha, \beta)]^{-1} \sum_0^\infty \left[\frac{g(\alpha, \beta) - g(z, w)}{g(\alpha, \beta)} \right]^n.$$

3.8. Suppose that f is holomorphic on O and that g, h are holomorphic on Ω open $\subset \mathbb{C}^2$ and satisfy

$$(g(z, w), h(z, w)) \in O, \quad (z, w) \in \Omega.$$

Show that $f[g(z, w), h(z, w)]$ with domain Ω is holomorphic on Ω .

3.9. Let $f: O \rightarrow \mathbb{C}$. Hartogs has shown that if f possesses first partial derivatives at each point of O , then f is holomorphic on O . This theorem of course extends the one-variable theory in an admirable way. [We must keep in mind that the existence of the first partial derivatives without supplementary hypotheses does not take us very far in real analysis.] The proof involves aspects of real analysis not assumed in this book. For a treatment of the theorem cf. [21]. With our present resources we can prove a weaker theorem of the Hartogs type: *If f satisfies the Hartogs hypothesis and, in addition, for each compact $K \subset O$ we have*

$$\sup\{|f(z, w)|: (z, w) \in K\} < +\infty,$$

then f is holomorphic on O . Thus the Hartogs condition and continuity (or differentiability) on O guarantee that f is holomorphic on O .

Carry out a proof of the above weak Hartogs theorem along the following lines. Let $\overline{\Delta(\alpha; r)} \times \overline{\Delta(\beta; s)} \subset O$, r, s positive. We have

$$f(z + \alpha, w + \beta) = \sum_0^\infty g_k(w) z^k, \quad |z| < r, \quad |w| < s,$$

and

$$|g_k(w)| \leq M r^{-k}, \quad |w| < s, \quad k \in \mathbb{N},$$

for some positive M . It suffices to show, thanks to the Cauchy inequalities, that each

g_k is analytic on $\Delta(0; s)$ —a fact that may be proved inductively. It is clear for g_0 . Considering

$$\left[f(z + \alpha, w + \beta) - \sum_0^n g_k(w) z^k \right] z^{-(n+1)}$$

we see that, for each $z \in \Delta(0; r) - \{0\}$, $\sum_0^\infty g_{n+1+k}(w) z^k$ with domain $\Delta(0; s)$ is analytic on $\Delta(0; s)$ by virtue of the induction hypothesis. Complete the proof by estimating the size of

$$g_{n+1}(w) - \sum_0^\infty g_{n+1+k}(w) z^k$$

for small z .

4. CAUCHY'S MAJORANT CALCULUS

The object here is to systematize the method of undetermined coefficients for existence theorems. In certain classes of problems the *hypothesis* of the local existence of an *analytic* solution leads to the determination of its Taylor coefficients. However, there remains to be decided whether the sequence so obtained is the sequence of the coefficients of a power series having a nonzero radius of convergence whose sum composed with the appropriate translation yields *locally* the desired solution. The idea of Cauchy is to put in juxtaposition with the given problem a cognate problem where the data are furnished by analytic (holomorphic) functions and the entering Taylor coefficients are *nonnegative* and dominate the moduli of the corresponding coefficients associated with the matching data of the original problem.† The program is carried out by showing: (a) the Taylor coefficients of a solution for the given problem analytic at the point in question are determined; (b) if the cognate problem has a solution, so does the original problem; (c) a choice can be made of an allowed cognate problem for which there is an analytic solution.

Inevitably, the above formulation is vague. An examination of some concrete situations will reveal the fecundity of Cauchy's majorant calculus.

We consider first a classical existence and uniqueness problem for differential equations. Let A be holomorphic at each point of a closed bicylinder $\overline{\Delta(0; r)} \times \overline{\Delta(0; s)}$, $0 < r, s < +\infty$, so that

† In the two-variable theory the a_{mn} entering in

$$f(z + \alpha, w + \beta) = \sum_{(m,n) \in \mathbb{N}^2} a_{mn} z^m w^n$$

are the *Taylor coefficients* of f at (α, β) and, of course, have an interpretation in terms of differentiation.

$$A(z, w) = \sum_{(m,n) \in \mathbb{N}^2} a_{mn} z^m w^n$$

for the (z, w) in the given closed bicylinder. Let $\beta \in \Delta(0; s)$. We ask:

Does there exist U analytic on some disk $\Delta(0; \rho)$, $0 < \rho < r$, such that

$$U(0) = \beta, \quad U[\Delta(0; \rho)] \subset \Delta(0; s),$$

and

$$U'(z) = A[z, U(z)], \quad |z| < \rho? \quad (4.1)$$

We carry out the program of showing that the following statements hold: (a) All such U have the same Taylor coefficients at 0. (b) Let B be holomorphic on $\Delta(0; r) \times \Delta(0; s)$,

$$B(z, w) = \sum_{(m,n) \in \mathbb{N}^2} b_{mn} z^m w^n, \quad |a_{mn}| \leq b_{mn},$$

and let there exist V analytic on $\Delta(0; \rho)$, $0 < \rho < r$, such that

$$V(0) \geq |\beta|, \quad V[\Delta(0; \rho)] \subset \Delta(0; s),$$

$$V'(z) = B[z, V(z)], \quad |z| < \rho.$$

Then there exists U with domain $\Delta(0; \rho)$ having the above-stated properties. (c) Such a choice of B is available. We shall even go a step further and show, in addition: (d) Given $0 < d_0 < s$, there exists ρ , $0 < \rho < r$, such that for each $\beta \in \Delta(0; d_0)$ there exists U_β with domain $\Delta(0; \rho)$ satisfying the above-stated conditions and such that the function assigning to $U_\beta(z)$ to

$$(z, \beta) \in \Delta(0; \rho) \times \Delta(0; d_0)$$

is holomorphic on its domain.

(a) Suppose that $U(z) = \sum_0^\infty c_k z^k$, $|z| < \rho$, and that

$$U[\Delta(0; \rho)] \subset \Delta(0; s).$$

The chain rule is available for the calculation of the successive derivatives of $A[z, U(z)]$ as we readily deduce with the aid of Ex. 3.1 of the preceding section. We find with its use that

$$A[z, U(z)] = \sum_0^\infty P_n(c_0, \dots, c_n) z^n,$$

where when $n \geq 1$ the coefficient $P_n(c_0, \dots, c_n)$ is the value at $(c_1, \dots, c_n)$ of a polynomial whose coefficients are the values at c_0 of functions analytic in $\Delta(0; s)$ which are independent of allowed U . Hence if U is a solution of (4.1)

satisfying $U(0) = \beta$, we have

$$\begin{aligned} c_0 &= \beta, \\ (n+1)c_{n+1} &= P_n(c_0, \dots, c_n), \quad n \in \mathbb{N}. \end{aligned} \quad (4.2)$$

It follows that all U fulfilling the originally stipulated conditions (ending in (4.1)) have the same Taylor coefficients at 0.

(b) Given an admitted B and $V(z) = \sum_0^\infty d_k z^k$ we see, paraphrasing the above argument leading to (4.2), that

$$B[z, V(z)] = \sum_0^\infty Q_n(d_0, \dots, d_n) z^n.$$

Here the remark made above concerning P_n is pertinent, *mutatis mutandis*. Noting that $b_{mn} \geq 0$ we see that when $n \geq 1$ the coefficient $Q_n(d_0, \dots, d_n)$ is the value at $(d_1, \dots, d_n)$ of a polynomial with nonnegative coefficients. From the fact that V satisfies the stated differential equation we have, as in (4.2),

$$(n+1)d_{n+1} = Q_n(d_0, \dots, d_n). \quad (4.3)$$

Comparing the coefficients of P_n and Q_n , we see that, if $(\alpha_1, \dots, \alpha_n) \in \mathbb{C}^n$ satisfies $|\alpha_k| \leq \gamma_k$, $k = 1, \dots, n$, then

$$|P_n(\beta, \alpha_1, \dots, \alpha_n)| \leq Q_n(d_0, \gamma_1, \dots, \gamma_n), \quad n \geq 1. \quad (4.4)$$

From (4.3) we conclude inductively that $d_n \geq 0$, $n \in \mathbb{N}$, and from (4.4) that the solution (c_n) of the system (4.2) satisfies $|c_n| \leq d_n$, $n \in \mathbb{N}$. The series $\sum_0^\infty c_k z^k$ is convergent for $|z| < \rho$; U , its sum restricted to $\Delta(0; \rho)$, satisfies $|U(z)| \leq V(|z|) < s$, $|z| < \rho$, and by (4.2) we conclude that U indeed satisfies (4.1).

Since (4.2) yields that c_n is the value at β of an analytic function, $|\beta| < s$, the uniform convergence of $\sum_0^\infty c_n(\beta) z^n$ in $\Delta(0; \rho) \times \Delta(0; d_0)$ shows that the function assigning $U_\beta(z)$ to $(z, \beta) \in \Delta(0; \rho) \times \Delta(0; d_0)$ is holomorphic on its domain. The assertion (d) is a consequence, once the cognate B , V are produced.

(c) Let

$$M = \max\{|A(z, w)| : |z| \leq r, |w| \leq s\}.$$

We propose

$$B(z, w) = \sum_{(m, n) \in \mathbb{N}^2} M \left(\frac{z}{r} \right)^m \left(\frac{w}{s} \right)^n = M \left(1 - \frac{z}{r} \right)^{-1} \left(1 - \frac{w}{s} \right)^{-1}. \quad (4.5)$$

To exhibit an allowed V , we obtain first a necessary condition on such V . Indeed, V satisfies

$$V(z) - \frac{[V(z)]^2}{2s} = -Mr \operatorname{Log}\left(1 - \frac{z}{r}\right) + \left(d_0 - \frac{d_0^2}{2s}\right), \quad (4.6)$$

$|z| < \rho$, as we see by integration. We are led to construct an allowed V as follows. We observe that

$$z - \frac{z^2}{2s}$$

with domain $\Delta(0; s)$ is univalent and we introduce φ , its inverse.[†] We take ρ , $0 < \rho < r$, so small that the right-hand side of (4.6) takes values in the domain of φ , $|z| < \rho$, d_0 being given on $[0, s[$. We propose for V the composition of φ and the right-hand side of (4.6) with domain $\Delta(0; \rho)$. Clearly, V is analytic on $\Delta(0; \rho)$. So defined V satisfies (4.6) and $V(0) = d_0$. By its definition $V[\Delta(0; \rho)]$ is contained in $\Delta(0; s)$, the image of φ . Differentiating (4.6) we see that V is a solution of the differential equation in question. The verification of (c) is complete.

EXERCISE

4.1. Suppose that in the situation just studied the Taylor coefficients u_{mn} of $U_\beta(z)$ are such that $u_{mn} z^m \beta^n$ is summable to a sum of modulus less than s for $|z| < \lambda \leq r$, $|\beta| < \mu$. Let $S_\beta(z)$ denote the value of the sum at (z, β) . Show that

$$S_\beta'(z) = A[z, S_\beta(z)], \quad |z| < \lambda,$$

for each β satisfying $|\beta| < \mu$.

The Cauchy majorant calculus, which we have explicated in the case of the above existence problem, may be profitably used in numerous situations. An important example is furnished by the *implicit function theorem*. We continue with A as above, but further restricted to satisfy: $A(0, 0) = 0$, $A_w(0, 0) \neq 0$. We wish to obtain information on the solution set

$$\mathfrak{A} = \{(z, w): A(z, w) = 0\},$$

in particular, on the nature of $\mathfrak{A} \cap [\Delta(0; t) \times \Delta(0; u)]$ for t and u small. The implicit function theorem asserts that for A so specified there exist t, u satisfying $0 < t \leq r$, $0 < u \leq s$, such that

$$\mathfrak{A} \cap [\Delta(0; t) \times \Delta(0; u)] \quad (4.7)$$

is a function analytic on $\Delta(0; t)$. Naturally, we are referring to the complex version of the theorem. The real variable form of the theorem varies some-

[†] We could be very explicit about φ and represent it with analytic square roots. It is not necessary to do so.

what from this as one would expect on comparing differential analysis in the real domain with that in the complex domain.

We shall show that (4.7) enjoys the property of being a function for suitable small t and u and shall leave the study of the analyticity by means of the Cauchy majorant calculus as an exercise. We have, when $0 < t \leq r$, $0 < u \leq s$, the equality

$$A(z, w_2) - A(z, w_1) = \int_{w_1 w_2}^{\rightarrow} A_w(z, w) dw,$$

$|z| < t$, $|w_1|, |w_2| < u$, and hence, when t and u are so small that

$$|A_w(z, w) - a_{01}| < |a_{01}|/2$$

for $|z| < t$, $|w| < u$,

$$|A(z, w_2) - A(z, w_1)| \geq |a_{01}| |w_2 - w_1|/2.$$

The function property follows, but it is to be observed that we have not even shown that the domain of the function is a neighborhood of 0.

EXERCISES

4.2. Apply the Cauchy majorant calculus to establish the existence of a function U analytic at 0 and satisfying $U(0) = 0$ as well as $|U(z)| < s$ and

$$U(z) = \frac{1}{a_{01}} \{-A[z, U(z)] + a_{01} U(z)\}$$

for $|z| < \rho$ for some ρ , $0 < \rho < r$. Show that this result taken with the above remarks of the text yields the implicit function theorem.

4.3. Show that the objective of Ex. 4.2 may be attained by applying the existence theorem of the text for differential equations to obtain U analytic at 0, and satisfying $U(0) = 0$ and (with due regard for admissibility)

$$U'(z) = -\frac{A_z[z, U(z)]}{A_w[z, U(z)]}, \quad z \text{ small.}$$

4.4. Show that the implicit function theorem may also be proved by considering

$$\frac{1}{2\pi i} \oint_{C(0; u)} \frac{w A_w(z, w)}{A(z, w)} dw$$

for u suitably chosen and z small. Hint: Take u , $0 < u \leq s$, so that $A(0, w) \neq 0$, $0 < |w| \leq u$, and thereupon z so small that

$$|A(z, w) - A(0, w)| < |A(0, w)|, \quad |w| = u.$$

4.5. Treat the existence and uniqueness theorem for a system of the form

$$w_k' = A_k(z, w_1, \dots, w_n), \quad k = 1, \dots, n,$$

where each A_k is holomorphic on $[\Delta(0; r)]^{n+1}$. It is to be understood that “holomorphic” is defined in $\mathbb{C}^{n+1}$ by a direct paraphrase of the $\mathbb{C}^2$ definitions.

5. VECTOR-VALUED FUNCTIONS

Numerous aspects of the classical theory of analytic functions may be extended to the theory of analytic functions taking values in a complex Banach space—a notion that we shall presently define. Our object in considering briefly a theory of vector-valued analytic functions is to show how natural and useful the power series methods are. We shall see that the methods to be developed lead to a treatment of the existence theorems of systems of linear differential equations which is formally elegant and avoids a plethora of notation. For a thorough account of vector-valued analytic functions cf. [56]. For future reference we agree to state explicitly that we are dealing with vector-valued functions when this is the case.

We start with the notion of a right vector space over $\mathbb{C}$. Given a set V and maps $+: V^2 \rightarrow V$, $\cdot: V \times \mathbb{C} \rightarrow V$, we say that $(V, +, \cdot)$ is a *right vector space over $\mathbb{C}$* provided that $(V, +)$ is an abelian group and that $\cdot$, *right multiplication by scalars*, satisfies:

$$\begin{aligned}(x + y)\alpha &= x\alpha + y\alpha, & x(\alpha + \beta) &= x\alpha + x\beta, \\ x(\alpha\beta) &= (x\alpha)\beta, & x1 &= x,\end{aligned}$$

where $x, y \in V$, $\alpha, \beta \in \mathbb{C}$. Of course, we can introduce left vector spaces as well. Solely on notational grounds we shall consider the “right” theory. The usual practice of omitting $\cdot$ will be followed. Further, the earlier conventions made for algebraic and topological structures are to prevail, so that we term V itself a vector space. The elements of V are termed *vectors*. We shall denote by O the neutral element of V considered as an abelian group. A map of V into $\mathbb{R}$ —the image of $x \in V$ with respect to it will be denoted $\|x\|$ —is termed a *norm* provided that the following conditions are fulfilled:

- (a) If $\|x\| = 0$, then $x = O$.
- (b) $\|x\alpha\| = \|x\| |\alpha|$.
- (c) $\|x + y\| \leq \|x\| + \|y\|$.

Here $x, y \in V$ and $\alpha \in \mathbb{C}$. We term $\|x\|$ the *norm* of x . It is immediate from $00 = O$ and (b) that $\|O\| = 0$, and from this fact and (c) applied to $x + (-x)$ and (b), that $\|x\| \geq 0$. V is endowed with a metric space structure by $\rho(x, y) = \|y - x\|$. The composite $(V, +, \cdot, \|\cdot\|)$ is called a *normed (right) vector space over $\mathbb{C}$* . It is termed a (right) *complex Banach space* provided that (V, ρ) is a complete metric space. The designation “right” will be understood in what follows. For the remainder of this section V will be a complex Banach

space. We shall be especially concerned with maps from a subset of $\mathbf{C}$ into V .

By a *bounded endomorphism* of V is meant a map $T: V \rightarrow V$ which is *linear*, that is, satisfies

$$T(x + y) = T(x) + T(y), \quad T(x\alpha) = T(x)\alpha,$$

where $x, y \in V, \alpha \in \mathbf{C}$, and which also satisfies the condition

$$N(T) = \sup\{\|T(x)\|: \|x\| = 1\} < +\infty.$$

$N(T)$ is termed the *norm* of T . We have $\|T(x)\| \leq N(T)\|x\|$, $x \in V$, for a bounded endomorphism T as we see from considering

$$T(x) = T[x(\|x\|)^{-1}]\|x\|$$

when $x \neq 0$. We introduce $\mathbf{B}(V)$, the set of bounded endomorphisms of V , and observe that $\mathbf{B}(V)$ is itself a (right) complex Banach space with the following agreements. Given $S, T \in \mathbf{B}(V)$, $\alpha \in \mathbf{C}$, $S + T$ is the map of V into itself which takes $x \in V$ into $S(x) + T(x)$ and $S\alpha$ the map taking x into $S(x)\alpha$. In addition, N is proposed as a norm. It is routine to verify that with these agreements $\mathbf{B}(V)$ is a complex Banach space and that further $\mathbf{B}(V)$ is closed with respect to composition so that when $S, T \in \mathbf{B}(V)$ we have $S \circ T \in \mathbf{B}(V)$. Indeed, we have $N(S \circ T) \leq N(S)N(T)$.

EXERCISE

5.1. Furnish the details omitted in the above discussion.

We note that the usual notions of elementary real and complex differential and integral calculus may be introduced for functions taking values in V when the norm replaces absolute value. Thus, in particular, the notion of the derivative at a of a map f from a neighborhood (in the sense of the $\mathbf{R}$ - or $\mathbf{C}$ -topology) of a into V lies at hand.† The following lemma is very useful in the elementary theory of vector-valued functions.

Lemma 5.1: *Let $-\infty < a < b < +\infty$. Let $f: [a, b] \rightarrow V$. If $f'(t) = 0$, $a \leq t \leq b$, then f is constant.*

PROOF: We introduce $g(t) = \|f(t) - f(a)\|$ and note that

$$|g(s) - g(t)| \leq \|f(s) - f(t)\|.$$

It follows that g is the constant zero and consequently f is the constant $f(a)$.

† We introduce $[f(x) - f(a)](x - a)^{-1}$ and take the limit at a .

EXERCISE

5.2. Extend the developments of Ch. V to construct a Cauchy theory for functions taking values in a complex Banach space, the starting point of the discussion being the study of Riemann–Stieltjes integrals $\int_a^b f dg$ where f takes values in V and g in $\mathbb{C}$. Lemma 5.1 will be seen to play the same important role as its cognate in the theory of complex-valued functions.

In what follows we shall concentrate on the power series aspects of vector-valued analytic functions. We suppose that $(a_n): \mathbb{N} \rightarrow V$. We are led to a paraphrase of the theory of power series for $\sum_0^\infty a_k z^k$. Here we consider complex z . Among the extensions that are automatic are the property of radius of convergence encountered at the outset in Ch. II, the Cauchy–Hadamard theorem, the identity theorem, and term-by-term differentiation. It is to be noted that if $(b_n): \mathbb{N} \rightarrow \mathbf{B}(V)$ and $\sum_0^\infty a_k z^k$ and $\sum_0^\infty b_k z^k$ converge on $\Delta(0; r)$, $r > 0$, having, respectively, sums A and B on $\Delta(0; r)$, then

$$B(z) [A(z)] = \sum_0^\infty c_n z^n, \quad |z| < r,$$

where

$$c_n = \sum_{k+j=n} b_k(a_j), \quad n \in \mathbb{N}$$

(Cauchy “multiplication”). The asserted equality may be established by appeal to the fact that $\sum_0^\infty \|a_k\| |z|^k$ and $\sum_0^\infty N(b_k) |z|^k$ are convergent for $|z| < r$ and by considering the limiting behavior of

$$\left(\sum_0^n b_k z^k \right) \left(\sum_0^n a_k z^k \right) - \sum_0^{2n} c_k z^k.$$

Account is to be taken of the fact that when $|z| < r$, $\lim_{n \rightarrow \infty} \|c_n z^n\| = 0$.

EXERCISE

5.3. Let A be an open subset of $\mathbb{C}$, let $a \in A$ and let $f: A \rightarrow V$ be of power series type. Let

$$f(a+z) = \sum_0^\infty c_k z^k$$

for small z . Show that the radius of convergence of $\sum_0^\infty c_k z^k$ is at least $d = \text{dist}(\{a\}, \mathbb{C} - A)$ and that its sum on $\Delta(0; d)$ is $f(a+z)$. The present exercise

extends Ex. 6.7, Ch. IV. Show that the method indicated in that exercise is available in the present setting with appropriate changes, principally in the question of treating differentiation under the integral sign.

Geometric series. Let I denote the identity map of V onto itself. It is trivial that $I \in \mathbf{B}(V)$. Let $b \in \mathbf{B}(V)$. We shall show the identity

$$(I - bz) \circ \left(\sum_0^{\infty} b^k z^k \right) = I = \left(\sum_0^{\infty} b^k z^k \right) \circ (I - bz), \quad |z| < [N(b)]^{-1}. \quad (5.1)$$

Here b^k denotes the k th iterate of b ($b^0 = I$, $b^{k+1} = b \circ b^k$). Indeed, it is easy to see that

$$(I - bz) \circ \left(\sum_0^n b^k z^k \right) = I - b^{n+1} z^{n+1} = \left(\sum_0^n b^k z^k \right) \circ (I - bz).$$

The identity (5.1) follows on taking the limit. The first half of the identity (5.1) shows that $I - bz$ maps V onto itself and the second half shows that $I - bz$ is univalent where in each instance $|z| < [N(b)]^{-1}$. Further under this condition $(I - bz)^{-1}$, the map of V onto itself inverse to $I - bz$, is equal to $\sum_0^{\infty} b^k z^k$.

We continue with $b \in \mathbf{B}(V)$ and introduce $E_b: \mathbf{C} \rightarrow \mathbf{B}(V)$ given by

$$E_b(z) = \sum_0^{\infty} b^k \frac{z^k}{k!}.$$

Clearly $E_b(0) = I$ and $E_b'(z) = b \circ E_b(z) = E_b(z) \circ b$. With

$$\varphi(z) = E_b(z) \circ E_b(-z) \quad \text{and} \quad \psi(z) = E_b(-z) \circ E_b(z)$$

we have the equality

$$\varphi'(z) = O_{\mathbf{B}} = \psi'(z), \quad z \in \mathbf{C},$$

where $O_{\mathbf{B}}$ is the neutral element of $\mathbf{B}(V)$. Hence we conclude that

$$E_b(z) \circ E_b(-z) = I = E_b(-z) \circ E_b(z) \quad (5.2)$$

and see that for each $z \in \mathbf{C}$ the endomorphism $E_b(z)$ is *invertible*, that is, it is a member of $\mathbf{B}(V)$ which is a univalent map of V onto itself whose inverse is also a member of $\mathbf{B}(V)$.† Suppose that $a \in V$ and that we seek $y: \mathbf{C} \rightarrow V$ satisfying $y(0) = a$ and $y'(z) = b[y(z)]$, $z \in \mathbf{C}$. If y does satisfy these conditions, we see that $E_b(-z)[y(z)]$ with domain $\mathbf{C}$ has a derivative taking the value O

† We are not presupposing a knowledge of the theorem of Banach space theory which states that a bounded univalent linear transformation of a Banach space onto a Banach space has a bounded inverse.

at each point of $\mathbf{C}$. Hence $E_b(-z)[y(z)] = a$ and we conclude with the aid of (5.2) that $y(z) = E_b(z)(a)$. In the opposite direction, it is immediate that $E_b(z)(a)$ fulfills both of the imposed conditions. It is the unique solution of the proposed problem.

EXERCISE

5.4. Develop the preceding discussion with particular attention to the formulas for the derivative of $f(z)[g(z)]$ where g takes values in V and f in $\mathbf{B}(V)$ and also for $f(z) \circ g(z)$ where now both f and g take values in $\mathbf{B}(V)$. Suitable hypotheses insuring the validity of the formulas in question are to be given.

Let us now see how power series methods for vector-valued functions are applicable to the study of the theory of linear systems of differential equations. We shall start with the problem in its original setting and thereupon shall subsume it under a problem stated in vector terms. Let n be a positive integer. Let a_{jk} and c_j be analytic on $\Delta(0; r)$, where $r > 0$, $j, k = 1, \dots, n$. We wish to study the existence and uniqueness of functions $w_1, \dots, w_n$ with domain $\Delta(0; r)$ satisfying $w_j'(0) = \alpha_j$, $j = 1, \dots, n$, where the α_j are assigned complex numbers, and also

$$w_j' = \sum_{k=1}^n a_{jk} w_k + c_j, \quad j = 1, \dots, n. \quad (5.3)$$

Of course, the problem just stated, which is *not* settled by our earlier local study of §4, this chapter, without further considerations, can be treated in the same setting in which it is formulated; but the question may be reformulated in the language of a suitable Banach space. We shall see that a twofold advantage will be gained. Not only will the formalism be very simple but also the class of problems rendered accessible to the methods to be used will include certain problems involving infinite systems of differential equations. Indeed, let $V = \mathbf{C}^n$. Let $+$ be given by componentwise addition so that if $u = (u_1, \dots, u_n)$ and $v = (v_1, \dots, v_n)$ are members of $\mathbf{C}^n$, then $u + v$ is defined as that member of $\mathbf{C}^n$ whose j th component is $u_j + v_j$, $j = 1, \dots, n$. Let scalar multiplication be defined also by componentwise multiplication so that, with u as above and $\alpha \in \mathbf{C}$, we define αu as that member of $\mathbf{C}^n$ whose j th component is $u_j \alpha$, $j = 1, \dots, n$. We introduce a norm by

$$\|u\| = \left(\sum_{k=1}^n |u_k|^2 \right)^{1/2}$$

With these agreements we see that V is a complex Banach space. Let $A(z)$

denote the map taking $u \in V$ into the element of V whose j th component is

$$\sum_1^n a_{jk}(z) u_k,$$

$j = 1, \dots, n$. $A(z)$ is a bounded endomorphism of V . Let $C(z)$ denote the member of V whose j th component is $c_j(z)$, $j = 1, \dots, n$. The problem stated above is equivalent to investigating the existence and uniqueness of $W: \Delta(0; r) \rightarrow V$ satisfying $W(0) = (\alpha_1, \dots, \alpha_n)$ and the differential equation

$$W'(z) = A(z) [W(z)] + C(z), \quad |z| < r. \quad (5.4)$$

It is to be observed that A and C are vector-valued functions given as sums of power series on $\Delta(0; r)$. Here A takes values in $\mathbf{B}(V)$ and C in V .

Let us now put aside the restrictions of finiteness and the special nature of the Banach space under consideration. We suppose then that we have at hand an arbitrary complex Banach space V and A and C , sums of power series on $\Delta(0; r)$, the coefficients, a_n , of the first being in $\mathbf{B}(V)$ and the coefficients, c_n , of the second being in V , and that $w_0 \in V$. We seek to study $W: \Delta(0; r) \rightarrow V$ satisfying (5.4) and the initial condition $W(0) = w_0$. We see from the availability of the Cauchy theory that the problem reduces to the study of solutions which are sums of power series on $\Delta(0; r)$. If a solution $\sum_0^\infty b_k z^k$ is available, then by the standard arguments (term-by-term differentiation, Cauchy "multiplication" given earlier in this section, the identity theorem) we obtain

$$\begin{aligned} b_0 &= w_0, \\ b_{n+1}(n+1) &= \left[\sum_{j+k=n} a_j(b_k) \right] + c_n, \quad n \in \mathbf{N}. \end{aligned} \quad (5.5)$$

It follows from (5.5) that the problem has at most one solution. We note that if $\sum_0^\infty b_k z^k$ has radius of convergence at least r , where $(b_k): \mathbf{N} \rightarrow V$ is the unique sequence satisfying (5.5), then W , its sum on $\Delta(0; r)$, is a solution of our problem, for the initial condition is assured by the first equation and the remaining equations imply that $W'(z)$ and $A(z) [W(z)] + C(z)$ are the sum of the same series, $|z| < r$.

To control the radius of convergence of $\sum_0^\infty b_k z^k$ we proceed as follows. Let $0 < s < t < r$, and let

$$\mu_n = \max \{ \|b_k\| s^k : k = 0, \dots, n \},$$

$n \in \mathbf{N}$. It suffices to show that (μ_n) is bounded for then $\sum_0^\infty \|b_k\| \rho^k$ converges when $0 \leq \rho < s$ and hence, given the arbitrariness of s , when $0 \leq \rho < r$. The convergence of $\sum_0^\infty b_k z^k$ for $|z| < r$ follows. From (5.5) we obtain

$$(n+1) \|b_{n+1}\| s^{n+1} \leq s \left[\sum_{j+k=n} N(a_j) s^j \|b_k\| s^k \right] + s(\|c_n\| s^n).$$

Since both $\sum_0^\infty N(a_k) t^k$ and $\sum_0^\infty \|c_k\| t^k$ are convergent, we conclude that there exists α positive such that

$$\|b_{n+1}\| s^{n+1} \leq \frac{1}{n+1} \frac{\alpha}{1-s/t} \mu_n + \frac{\alpha}{n+1} \left(\frac{s}{t}\right)^n. \quad (5.6)$$

We put aside the trivial case where all μ_n are zero and notice that by (5.6) for n sufficiently large $\|b_{n+1}\| s^{n+1} < \mu_n$ and consequently $\mu_{n+1} = \mu_n$. The boundedness of (μ_n) follows.

EXERCISE

5.5. With A as above apply the method of undetermined coefficients to study the differential equations (a) $S'(z) = A(z) \circ S(z)$ and (b) $T'(z) = T(z) \circ A(z)$ with solutions taking values in $\mathbf{B}(V)$.* [Show that if $S(0) = I$, then $S(z)$ is invertible. Show that with S normalized by $S(0) = I$ the totality of solutions of (a) is given by $S(z) \circ C$, $C \in \mathbf{B}(V)$, and that the solution of $W'(z) = A(z) [W(z)]$ (taking values in V) satisfying $W(0) = w_0$ is $S(z)(w_0)$. Discuss the totality of the solutions of (b).]

An important application of the method of undetermined coefficients is to the following problem. Suppose that A is as above and that we are interested in the differential equation

$$W'(z) z = A(z) [W(z)]. \quad (5.7)$$

Here W is to take values in V . We have deliberately left open the question of what the domain of the solution is to be. The reason will become clear below. While problems of this kind have been studied exhaustively in the classical case where $V = \mathbf{C}^n$ (cf. [13]), much remains to be done in the setting of general complex Banach spaces. Initial work in this direction is due to E. Hille (*Proc. Internat. Symp. Linear Spaces*, Jerusalem, 1960, 263–273).

We continue under the supposition that $a_0 \neq O_{\mathbf{B}}$. If $\sum_0^\infty b_k z^k$ defines a solution of (5.7) on $\Delta(0; r)$, then we obtain

$$a_0(b_0) = O, \\ [I(n+1) - a_0](b_{n+1}) = \sum_0^n a_{n+1-j}(b_j), \quad n \in \mathbf{N}.$$

If $In - a_0$ were univalent for each $n \in \mathbf{N}$, the only solution analytic on $\Delta(0; r)$ would be the constant O . We put this approach aside. Instead, we ask whether by a suitable choice of a complex number λ we may find a solution of

(5.7) with domain a region $\Omega \subset \Delta(0; r) - \{0\}$, where the solution is of the form $U(z)s(z)$ where $U: \Omega \rightarrow V$ is analytic but not the constant O and s is an analytic λ th power of the identity map on Ω . Clearly it follows from (5.7) that the following condition is imposed: λ, Ω, U must be such that

$$U'(z)z + U(z)\lambda = A(z)[U(z)], \quad z \in \Omega. \quad (5.8)$$

Of course s of the stated type must exist. A converse proposition is also seen to hold. These observations lead to the following construction of a nontrivial solution under *favorable* circumstances (present in the $\mathbf{C}^n$ theory).

Suppose that $\lambda \in \mathbf{C}$ is such that $I\lambda - a_0$ is not univalent and that

$$I[\lambda + (n+1)] - a_0$$

maps V onto itself, $n \in \mathbf{N}$. Let $b \in V - \{O\}$ satisfy $(I\lambda - a_0)(b) = O$ and let (b_n) be a sequence of vectors in V satisfying

$$b_0 = b, \quad (5.9)$$

$$\{I[\lambda + (n+1)] - a_0\}(b_{n+1}) = \sum_0^n a_{n+1-j}(b_j), \quad n \in \mathbf{N}.$$

Then $\sum_0^\infty b_k z^k$ has radius of convergence $\geq r$. Further if Ω is any subregion of $\Delta(0; r) - \{0\}$ on which there exists an analytic λ th power of the identity map of Ω onto itself, say s , and U is the sum of $\sum_0^\infty b_k z^k$ on Ω , then $U(z)s(z)$ satisfies (5.7) at each point of Ω .

EXERCISES

5.6. Prove the above result. It is convenient to treat first the radius of convergence. The argument used for the system (5.5) may be paralleled here. The fact that (5.9) is satisfied together with the condition on the radius of convergence yields the result that the differential equation of (5.8) is satisfied on $\Delta(0; r)$.

5.7. Show that Bessel's equation

$$z^2 u''(z) + z u'(z) + (z^2 - \lambda^2) u(z) = 0,$$

$\lambda \in \mathbf{C}$, is equivalent to the system

$$\begin{aligned} z u'(z) &= v(z), \\ z v'(z) &= (\lambda^2 - z^2) u(z), \end{aligned} \quad (5.10)$$

in the sense that the solutions of Bessel's equation are first components of the solutions of (5.10) and vice versa, the domains being open subsets of $\mathbf{C}$. Show that the method of the present section leads to solutions of

$$z w''(z) + (2\lambda + 1) w'(z) + z w(z) = 0$$

which turn out to be restrictions of entire functions when $\operatorname{Re} \lambda \geq 0$ (cf. Ex. 2.13, Ch. VI).

5.8. Examine the hypergeometric equation of Ex. 3.12, Ch. IV, from the present point of view at 0 and 1.

6. ANALYTIC CONTINUATION

The notion of analytic continuation, which will be introduced in this section, gives complex function theory a special sense of coherence. We shall approach the subject twice, first in the present chapter, where the concept of a function element is used, and later in Ch. XVIII, where the problem is viewed as a part of the modern theory of Riemann surfaces and the advantageous alternative formulation of H. Weyl is central. The less sophisticated approach of the present section has the merit of being very accessible as well as useful in applications. We shall take care in Ch. XVIII to study the relation between the two approaches.

In the exposition that follows we consider complex-valued functions.

We consider ordered pairs (a, P) where $a \in \mathbb{C}$ and P is of the form $S \circ \tau$, where S is the sum of a power series with radius of convergence $r > 0$ on the interior of its convergence set and τ is $z - a$ with domain $\Delta(a; r)$.† By an *analytic element* (or, more simply, an *element*) is meant such an ordered pair (a, P) . An element (b, Q) is termed a *direct continuation* of (a, P) provided that b is in the domain of P and $Q(z) = P(z)$ for z near b (and hence for z in the intersection of the domains of P and Q). Given an element (a, P) and b in the domain of P there is a unique element (b, Q) which is a direct continuation of (a, P) . The uniqueness is obvious by the identity theorem for power series. Further (b, Q) is a direct continuation of (a, P) where Q is simply the sum of

$$\sum_{k=0}^{\infty} \frac{P^{(k)}(b)}{k!} (z - b)^k$$

on the interior of its convergence set. It is to be noted that the existence of a direct continuation may be considered in purely Weierstrassian terms, that is, exclusively in terms of power series concepts without reference to the Cauchy theory. It suffices to use such ideas as sums of functions and the identity theorem for power series.

† It must be emphasized that if P is entire and $a \neq b$, then (a, P) and (b, P) are distinct since they are ordered pairs with distinct first components. At this point it may seem that we are laboring the obvious, but the fact is essential when we turn to incorporating these notions into the theory of Riemann surfaces. It is only the entire P which necessitate the "ordered pair" approach.

EXERCISE

6.1. Develop these remarks.

We denote the component Q of the direct continuation (b, Q) of (a, P) by P_b and the radius of convergence of the power series whose sum is $P_b(b + z)$, z small, by $r(b)$. Clearly if $r(a) = +\infty$, then $r(b) = +\infty$. If $r(a) < +\infty$, we have the following inequality:

$$r(a) - |b - a| \leq r(b) \leq r(a) + |b - a|. \quad (6.1)$$

The left inequality is immediate. If $r(b) > r(a) + |b - a|$, then we would conclude that $\Delta[a; r(a)] \subset \Delta[b; r(b)]$ and $P(z) = P_b(z)$ for z in $\Delta[a; r(a)]$, and the direct continuation of (b, P_b) with first component a , would have an associated radius (in the sense of the first sentence of this paragraph) exceeding $r(a)$. Contradiction. The right inequality follows.

Under *suitable* conditions we may use direct continuations of an element (a, P) to obtain a function f which is analytic on a region Ω containing the domain of P properly and which has the property that P is the restriction of f to the domain of P .† In any case we have

Theorem 6.1: $\bigcup P_b$ is a function, analytic on its domain, the union being taken over $b \in \Delta[a; r(a)]$.

PROOF: We put aside the obvious case where P is entire. It suffices to show that $P_b(z) = P_c(z)$ when z is in the intersection of the domains of P_b and P_c . We take advantage of the convexity property of circular disks. We note that

$$A = \Delta[b; r(b)] \cap \Delta[c; r(c)]$$

is open and convex. Let $d \in A$. Since

$$|b - c| \leq |b - d| + |d - c| < r(b) + r(c),$$

we see that

$$\alpha = \frac{r(c)b + r(b)c}{r(b) + r(c)} \in A \cap \Delta[a; r(a)].$$

Now P and P_b agree in some neighborhood of α . The same is true of P and P_c and consequently the same is true of P_b and P_c . Consequently, $P_b(z) = P_c(z)$, $z \in A$. The theorem follows.

† For example, $a = 0$, $P(z) = (1 + z)^{-1}$.

Theorem 6.1 leads to the following important conclusion: If $r(a) < +\infty$, there exists $w \in C[a; r(a)]$ such that

$$r(z) \leq |z - w|, \quad z \in \Delta[a; r(a)]. \quad (6.2)$$

Continuing with the case where $r(a) < +\infty$, we say that $w \in C[a; r(a)]$ is a *singularity of P* provided that there exists no f analytic at each point of $\Delta[a; r(a)] \cup \{w\}$ whose restriction to $\Delta[a; r(a)]$ is P . It is immediate that if w is a singularity of P , then (6.2) holds relative to w . The converse is true. For if w is not a singularity of P , and (w, Q) is the element such that $Q(z) = P(z)$ for z in the domain of P near w and $\Delta(w; \rho)$ is in the domain of Q , then

$$r(b) \geq \rho - |b - w| > |b - w|$$

for b near w . The inequality (6.2) is thereby violated.

EXERCISE

6.2. Suppose that $(a_k): \mathbf{N} \rightarrow \mathbf{R}^+$ and that $\sum_0^\infty a_k z^k$ has radius of convergence ρ , $0 < \rho < +\infty$. Let P denote the sum on $\Delta(0; \rho)$ and consider the direct continuations of $(0, P)$. Show that ρ is a singularity of P (Vivanti-Pringsheim) and that $r(z) \geq r(|z|) = \rho - |z|$. Show that this situation may occur even when the power series converges uniformly on $\Delta(0; \rho)$ by considering $\sum_0^\infty (k+1)^{-2} z^k$.

Continuation along a path. Let γ be a path in $\mathbf{C}$ with domain $[\alpha, \beta]$. A map π from $[\alpha, \beta]$ into the set of elements is termed a *continuation along γ* provided that for each $t \in [\alpha, \beta]$ the first component of $\pi(t)$ is $\gamma(t)$ and that for each such t for $s \in [\alpha, \beta]$ sufficiently near t the element $\pi(s)$ is a direct continuation of $\pi(t)$. We say that π is a *continuation from (a, P) to (b, Q)* provided that $\pi(\alpha) = (a, P)$ and $\pi(\beta) = (b, Q)$.

If π_1 and π_2 are continuations along γ , then either $\pi_1 = \pi_2$ or else $\pi_1(t) \neq \pi_2(t)$ for all $t \in [\alpha, \beta]$. This assertion is readily established with the aid of a connectedness argument. Indeed, the t for which $\pi_1(t) = \pi_2(t)$ clearly constitute an open set in the relative topology of $[\alpha, \beta]$. However if $\pi_1(t) \neq \pi_2(t)$, the second components of $\pi_1(t)$ and $\pi_2(t)$ are different (though, to be sure, they may very well take the same value at $\gamma(t)$) so that for s near t either $\gamma(s) = \gamma(t)$ and hence $\pi_1(s) \neq \pi_2(s)$, or else $\gamma(s) \neq \gamma(t)$ and hence $\pi_1(s) \neq \pi_2(s)$ since the values of their second components at $\gamma(s)$ are different. Since the set of t for which $\pi_1(t) \neq \pi_2(t)$ is also open in the sense of the relative topology of $[\alpha, \beta]$ and $[\alpha, \beta]$ is connected, the asserted property of continuations along γ follows.

It is obvious now that given γ with initial point a and an element (a, P) there is at most one continuation π along γ satisfying $\pi(\alpha) = (a, P)$.

EXERCISES

6.3. Let π be a continuation along a path γ . Let $\Delta[\gamma(t); \rho(t)]$ be the domain of the second component of $\pi(t)$. Show that ρ is either the constant $+\infty$ or else $\rho(t) < +\infty$, $\alpha \leq t \leq \beta$, and that in the second case for each such t we have

$$|\rho(s) - \rho(t)| \leq |\gamma(s) - \gamma(t)|$$

for $s \in [\alpha, \beta]$ sufficiently near t .

6.4. Given an element (a, P) and a path γ in the domain of P . Show that there is a unique continuation π along γ satisfying

$$\pi(\alpha) = (\gamma(\alpha), P_{\gamma(\alpha)}).$$

6.5. Let π be a continuation along a path γ having domain $[\alpha, \beta]$ and let φ denote a continuous map of $[\alpha_1, \beta_1]$ into $[\alpha, \beta]$, $-\infty < \alpha_1 < \beta_1 < +\infty$. Show that $\pi \circ \varphi$ is a continuation along $\gamma \circ \varphi$.

It is easily verified that continuation along paths induces an equivalence relation in the set of elements. We say that (b, Q) is “*equivalent*” to (a, P) provided that there exists a continuation π from (a, P) to (b, Q) . It is routine to verify that the relation so specified among the elements is reflexive ($\pi(t) = (a, P)$, $0 \leq t \leq 1$), symmetric (consider $\pi(1-t)$), and transitive ($\pi(t) = \pi_1(2t)$, $0 \leq t \leq \frac{1}{2}$, $\pi(t) = \pi_2(2t-1)$, $\frac{1}{2} < t \leq 1$, where π_1 is a continuation from (a, P) to (b, Q) and π_2 is a continuation from (b, Q) to (c, R)). We are using here the standard domain $[0, 1]$ for continuations. The resulting equivalence classes will be termed *Weierstrass continuation classes* or more simply *Weierstrass classes*. We shall see in Ch. XVIII that Weierstrass classes may be endowed with a Riemann surface structure. It is easily comprehended that Weierstrass classes lack a desirable maximality—for example, poles and the point at ∞ are not taken into regard. Just how far short of maximality the Weierstrass classes fall will be considered in §2, Ch. XVIII. It will be seen that the “oversights” have an isolated character.

Given a Weierstrass class $\mathbf{W}$ we say that a path γ in $\mathbf{C}$ with domain, say $[\alpha, \beta]$, is a *path of singularity* for $\mathbf{W}$ provided that there exists $\pi: [\alpha, \beta[\rightarrow \mathbf{W}$ such that $\pi|[\alpha, \delta]$ is a continuation along $\gamma|[\alpha, \delta]$ for each $\delta \in]\alpha, \beta[$, but there does not exist a continuation π_1 along γ satisfying $\pi_1(\alpha) = \pi(\alpha)$.

EXERCISES

6.6. Using the notation ρ of Ex. 6.3, show that γ is a path of singularity for $\mathbf{W}$ if and only if there exists $\pi: [\alpha, \beta[\rightarrow \mathbf{W}$ such that $\pi|[\alpha, \delta]$ is a continuation along $\gamma|[\alpha, \delta]$ for each $\delta \in]\alpha, \beta[$ and $\lim_{\delta \rightarrow \beta} \rho = 0$.

6.7. Show that if $(0, P)$ is an element and the domain of P is $\Delta(0; r)$, $r < +\infty$, then $\vec{0}_w$ is a path of singularity for the Weierstrass class containing $(0, P)$ for at least one $w \in C(0; r)$.

6.8. Let f be analytic on a region $A \subset \mathbb{C}$. Show that the set of elements (a, P^a) , $a \in A$, satisfying $P^a(z) = f(z)$, z near a , belong to a Weierstrass class and that, in fact, if γ is a path in A , then $(\gamma(t), P^{\gamma(t)})$ is a continuation along γ .

Monodromy theorem. This theorem is an important instrument for constructing analytic functions associated with a given Weierstrass class. The topological background will have a familiar ring from the developments of Ch. V. We consider a homotopically simply connected region $A \subset \mathbb{C}$ and an element (a, P) where $a \in A$. It is assumed that for *all* paths γ in A with initial point a and domain $[0, 1]$ there exists a continuation π_γ along γ satisfying $\pi_\gamma(0) = (a, P)$. The monodromy theorem may be stated as follows.

Theorem 6.2: *Under the above hypotheses there exists a unique function f analytic on A such that $f(z) = P(z)$ for z near a .*

REMARK: It is to be emphasized that it would be a *flagrant* error to suppose that the hypothesis on the paths γ might be replaced by the weaker one that for each $b \in A$ there *exists* a path γ in A with initial point a and terminal point b along which there exists a continuation π satisfying $\pi(0) = (a, P)$ (cf. Ex. 6.12 below of this section).

PROOF OF THE MONODROMY THEOREM. We show that if γ and Γ are paths in A having initial point a and a common terminal point, then $\pi_\gamma(1) = \pi_\Gamma(1)$.† Granting that this is so, we propose for f the set of ordered pairs

$$(\gamma(1), P^\gamma[\gamma(1)]), \quad (6.3)$$

where P^γ is the second component of $\pi_\gamma(1)$, all admitted γ being taken into account. Clearly f is a function with domain A . That f is analytic on A may be seen by noting that for given γ we have for z in some disk centered at $\gamma(1)$: $f(z) = P^\gamma(z)$. This may be verified by using the path Γ satisfying

$$\begin{aligned} \Gamma(t) &= \gamma(2t), & 0 \leq t \leq \frac{1}{2}, \\ \Gamma(t) &= (2t - 1)z + (2 - 2t)\gamma(1), & \frac{1}{2} < t \leq 1. \end{aligned}$$

The uniqueness of f is immediate.

Let us turn to the proof of the equality

$$\pi_\gamma(1) = \pi_\Gamma(1).$$

We note, as was remarked by Koebe, that the domain of P^γ is of the form $\Delta[\gamma(1); r]$ where $r \geq \text{dist}(\{\gamma(1)\}, \mathbb{C} - A)$ since P^γ admits continuation along all $\gamma(1) \rightarrow w$ where $|w - \gamma(1)|$ is less than $\text{dist}(\{\gamma(1)\}, \mathbb{C} - A)$. For this reason we put

† This explains the terminology “monodromy” = “one course.”

aside the case where $A = \mathbb{C}$ since P is then entire. To continue, we introduce a continuous map φ of $[0, 1]^2$ into A satisfying: $\varphi(s, 0) = \gamma(s)$, $\varphi(s, 1) = \Gamma(s)$, $0 \leq s \leq 1$; $\varphi(0, t) = a$, $\varphi(1, t) = b$, $0 \leq t \leq 1$. Let m denote the distance from the image of φ to $\mathbb{C} - A$. We take n , a positive integer, so that $\omega(n^{-1}\sqrt{2}) < m$ where ω is the modulus of continuity of φ . For each $k \in \langle 0, n \rangle$ we define e_{jk} recursively by: $e_{0k} = (a, P)$, $e_{j+1,k}$ is the direct continuation of e_{jk} which has first component $\varphi[n^{-1}(j+1), n^{-1}k]$. Because of the observation made at the beginning of this paragraph the e_{jk} are well defined. We next note that for each $j \in \langle 0, n \rangle$ it is the case that $e_{j,k+1}$ is a direct continuation of e_{jk} , $k = 0, \dots, n-1$. This is trivial for $j = 0$. Suppose that the assertion holds for a given $j \in \langle 0, n-1 \rangle$. Because of the choice of n the points $\varphi[n^{-1}j, n^{-1}(k+1)]$ and $\varphi[n^{-1}(j+1), n^{-1}(k+1)]$ are in the disk of radius m having center $\varphi(n^{-1}j, n^{-1}k)$. Hence $e_{j+1,k+1}$ is a direct continuation of e_{jk} . Since $e_{j+1,k}$ is also a direct continuation of e_{jk} and the segment

$$[\varphi[n^{-1}(j+1), n^{-1}k], \varphi[n^{-1}(j+1), n^{-1}(k+1)]]$$

lies in the disk with radius m and center $\varphi(n^{-1}j, n^{-1}k)$, we conclude that $e_{j+1,k+1}$ is a direct continuation of $e_{j+1,k}$, $k = 0, \dots, n-1$ (cf. Ex. 6.8, this section). Our assertion follows. On specializing j to be n , we conclude $e_{n0} = e_{nn}$.

To complete the proof we note that thanks to the condition on m and the property remarked at the beginning of the preceding paragraph we have $\pi_\gamma(n^{-1}k) = e_{k0}$, $\pi_\Gamma(n^{-1}k) = e_{kn}$, $k = 0, \dots, n$, since $\pi_\gamma(t)$ is a direct continuation of $\pi_\gamma(n^{-1}k)$ and $\pi_\Gamma(t)$ is a direct continuation of $\pi_\Gamma(n^{-1}k)$, $k \leq nt \leq k+1$, for $k = 0, \dots, n-1$. It follows that $\pi_\gamma(1) = \pi_\Gamma(1)$, and the proof is completed.

In the exercises below we explore some aspects of the significance of analytic continuation for complex function theory.

EXERCISES

6.9. Let π_1 and π_2 denote continuations along a path γ (domain $[0, 1]$) with initial point a . Let $\pi_1(0) = (a, P)$ and $\pi_2(0) = (a, Q)$. We shall use the P' notation of the monodromy theorem. Let S denote a primitive of P . Show that there exists a continuation π along γ satisfying $\pi(0) = (a, P')$ and that $\pi(1) = (\gamma(1), (P')')$. Prove the corresponding result for (a, S) . Treat the analogous question for (a, T) where $T(z) = P(z) + Q(z)$, z near a , as well as the corresponding question for products and quotients.

6.10. Let F be holomorphic on an open set $O \subset \mathbb{C}^{n+2}$ where $n \in \mathbb{N}$. Suppose that π is a continuation along a path γ with domain $[0, 1]$. Let P_i denote the second component of $\pi(t)$. We assume that

$$(\gamma(t), P_i[\gamma(t)], P_i'[\gamma(t)], \dots, P_i^{(n)}[\gamma(t)]) \in O,$$

$0 \leq t \leq 1$, and that

$$F[z, P_0(z), P_0'(z), \dots, P_0^{(n)}(z)] = 0,$$

for z near $\gamma(0)$. Show that

$$F[z, P_1(z), P_1'(z), \dots, P_1^{(n)}(z)] = 0$$

for z near $\gamma(1)$ (permanence principle for differential equations).

6.11. Let f be entire and not constant. For each $a \in \mathbb{C}$ such that $f'(a) \neq 0$ let $(f(a), P_a)$ denote the unique element with first component $f(a)$ satisfying $P_a[f(z)] = z$, z near a . Show that the set of these elements is a Weierstrass class.

6.12. Let $f = z(z-1)^2$, $z \in \mathbb{C}$. Let $(0, P)$ denote the unique element with first component 0 satisfying $f[P(z)] = z$, z small. Show that if $\Delta(0; r)$ is the domain of P , then $r < +\infty$, and that for each $z \in \mathbb{C}$ there exists a continuation π along a path with initial point 0 and terminal point z which satisfies $\pi(0) = (0, P)$. Hint: The roots of $f'(z) = 0$ are $\frac{1}{3}$ and 1. Given $a \in \mathbb{C}$ there exists a path γ with initial point 0 not passing through $\frac{1}{3}$ or 1 such that $f[\gamma(1)] = a$. Show that there exists a continuation π along $f \circ \gamma$ satisfying $\pi(0) = (0, P)$. In fact $\pi(t)$ is the unique element $(f \circ \gamma(t), P_t)$ for which $P_t[f \circ \gamma(t)] = \gamma(t)$ and $f[P_t(z)] = z$, z near $f \circ \gamma(t)$.

This shows that the misstated hypothesis of the monodromy theorem does not assure the conclusion.

6.13. Show that the set of all elements (a, λ) where λ is an analytic logarithm of the identity map on $\Delta(a; |a|)$, $a \in \mathbb{C} - \{0\}$, is a Weierstrass class. Show that a corresponding statement holds for the set of elements (a, P) where P is an analytic α th power of the identity map on $\Delta(a; |a|)$, $a \in \mathbb{C} - \{0\}$. Here α is a fixed member of $\mathbb{C} - \mathbb{N}$.

6.14. Let $0 \leq r < s \leq +\infty$. Let $a \in \mathbb{C}$. Let A be the annulus $\{r < |z - a| < s\}$. Let $b \in A$ and suppose that (b, P) is an element with the property that for each path γ (with domain $[0, 1]$) in A with initial point b there exists a continuation π_γ along γ satisfying $\pi_\gamma(0) = (b, P)$. Let c be a logarithm of $b - a$. Let (c, Q) be an element satisfying $Q(z) = P(a + e^z)$ for z near c . Show that for each path Γ (with domain $[0, 1]$) in $\{\log r < \operatorname{Re} z < \log s\}$ with initial point c there exists a continuation π_Γ along Γ satisfying $\pi_\Gamma(0) = (c, Q)$. Let f be the analytic function on

$$\{\log r < \operatorname{Re} z < \log s\}$$

satisfying $f(z) = Q(z)$, z near c . Show that the totality of $\pi_\gamma(1)$ consists of the elements $(a + e^w, P_w)$ where $\log r < \operatorname{Re} w < \log s$ and P_w is uniquely specified by $P_w(a + e^z) = f(z)$, z near w .

A particularly important case arises when f is periodic with period $2\pi in$, n a positive integer. Here $f(z) = F(e^{z/n})$ where F is analytic on $\{r^{1/n} < |z| < s^{1/n}\}$. The $\pi_\gamma(1)$ are now given by $(a + w^n, P_w)$ where $r^{1/n} < |w| < s^{1/n}$ and $P_w(a + z^n) = F(z)$, z near w . Show that f is periodic with period $2\pi in$ if there exists a closed curve κ in A , $\kappa(0) = b$, $O(a; \kappa) = n$, such that there exists a continuation π along $\kappa| [0, 1]$ satisfying $\pi(0) = \pi(1) = (b, P)$. What can be said in the other direction?

Consider the case $r = 0$ and obtain a theorem of the Weierstrass type (for isolated singularities) when f has period $2\pi in$. Is there such a theorem when f is not periodic?

6.15. Let A be a simply connected region in $\mathbb{C}$ and let $a_0, \dots, a_n$ be analytic on A , $n \in \mathbb{N}$. Let

$$F(z, w) = w^{n+1} + \sum_0^n a_k(z) w^k.$$

Suppose that for each $z \in A$ the set $\{w: F(z, w) = 0\}$ has $n + 1$ distinct members (or equivalently $F(z, w) = F_w(z, w) = 0$ has no solution). Show with the aid of the monodromy theorem that there exist exactly $n + 1$ distinct functions analytic on A , say $W_1, \dots, W_{n+1}$, satisfying $F[z, W_k(z)] = 0$, $z \in A$, $k = 1, \dots, n + 1$.

6.16. Extend the theory of analytic continuation to vector-valued analytic functions of a single complex variable.